UX CASE STUDY · PERSONAL PROJECT

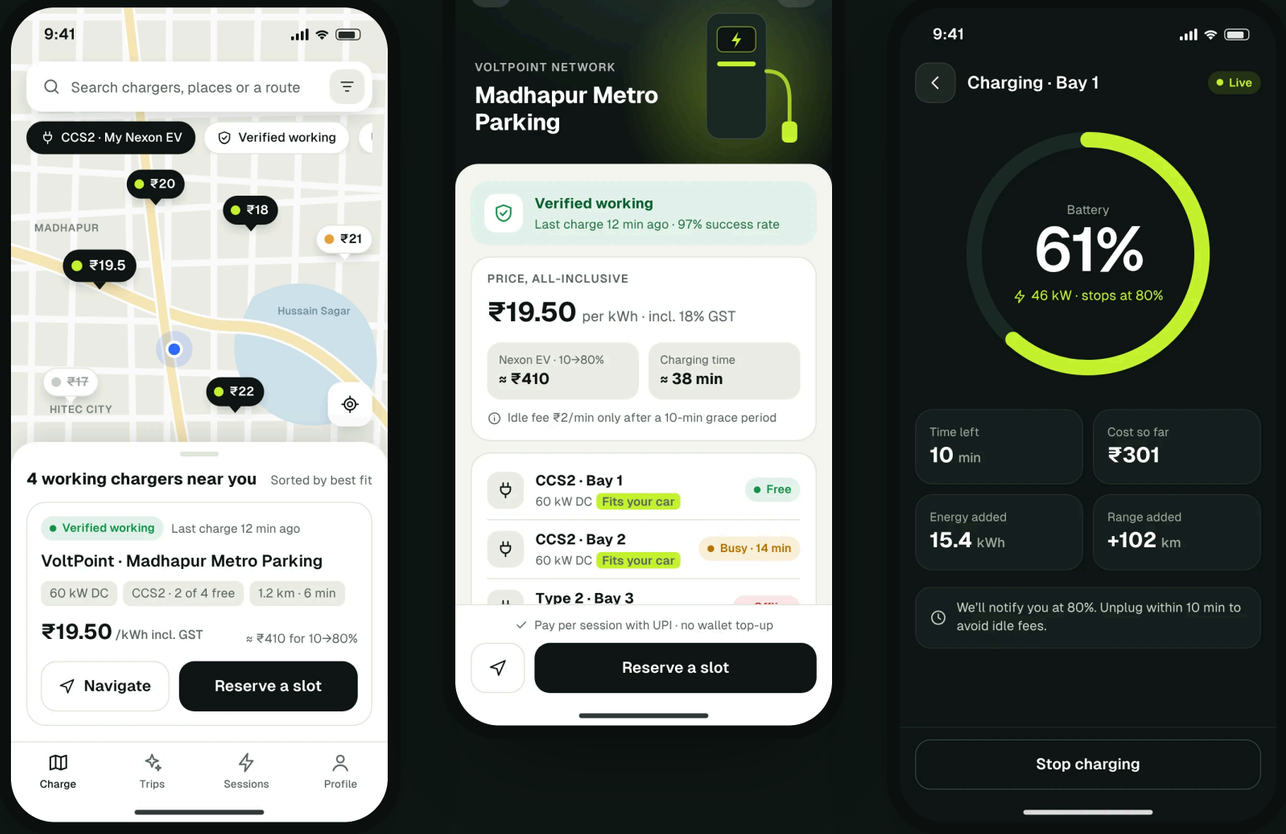
EVCS

An EV charging app for India that sends drivers only to chargers that actually work, shows the full price before they plug in, and lets them pay per session by UPI.

ROLE
Sole designer

PLATFORM
iOS and Android

TIMELINE
2019 · revisited 2026



Shiva Shanker Bantu · shivashankerbantu.github.io

From “where is a charger?” to “which one will actually work?”

THE PROBLEM

In 2019 there weren't enough chargers. By 2026 India has tens of thousands, but drivers can't trust them: chargers shown as available are broken, every network wants its own app and wallet, and the price only becomes clear after you plug in.

THE GOAL

One app that sends drivers only to chargers that work, shows the full price upfront, and lets them pay per session without wallets, with a clear plan for when a charger fails.

MY ROLE

Sole designer: research, personas, journey, user flow, app structure, visual identity, every screen and a clickable prototype. First version in 2019 (15 weeks), rebuilt from scratch in 2026.

What the research shows

2019: interviews with 10+ EV drivers. 2026: a review of the main charging apps in India, their app store reviews and published reports.

1 in 2

public charging stations in India were offline in August 2025.

Redseer, via EVDrive

17–20

apps a driver may juggle to find and pay for a working charger.

Tata Motors, via AckoDrive

₹100

typical minimum wallet top-up in each app before you can charge.

The Core

84%

of chargers checked in parts of Delhi weren't working, in a 2024 study.

IEEFA, via EV Infrastructure News

WHAT APP STORE REVIEWS KEEP REPEATING

- Sessions that fail to start
- Charging stops halfway
- Money stuck in wallets
- Slow refunds
- Support tickets open for weeks
- Price unclear until after charging

Process

01

Discover

2019

Interviews with 10+ EV drivers about charging, travel and apps

02

Define

2026

Live apps, app store reviews and published reports; personas

03

Ideate

Journey, user flow and app structure

04

Design

2026

New identity, mood board, 12 screens

05

Test

2019 · next

Tested with 5 users in 2019; prototype ready for drivers

Who it’s for

Built from the 2019 interviews and the 2026 research.

R

Ravi

CAB DRIVER, HYDERABAD

Drives 200 km a day for a ride-hailing app. Every hour lost at a broken charger is an hour without fares.

PAIN POINTS

- Drives to chargers the app says are free, and finds them broken
- Waits in queues without knowing for how long
- Has money stuck in several wallets

P

Priya

IT PROFESSIONAL

Lives in an apartment where she can’t install a charger, so she depends on public ones every week.

PAIN POINTS

- Only sees the real price after charging
- Sessions that fail halfway, with slow refunds
- Isn’t sure which connector fits her car

M

Mr. Rao

RETIRED, ROAD TRIPS

Drives to see family in other cities. Comfortable with UPI, but not with an account and wallet in every app.

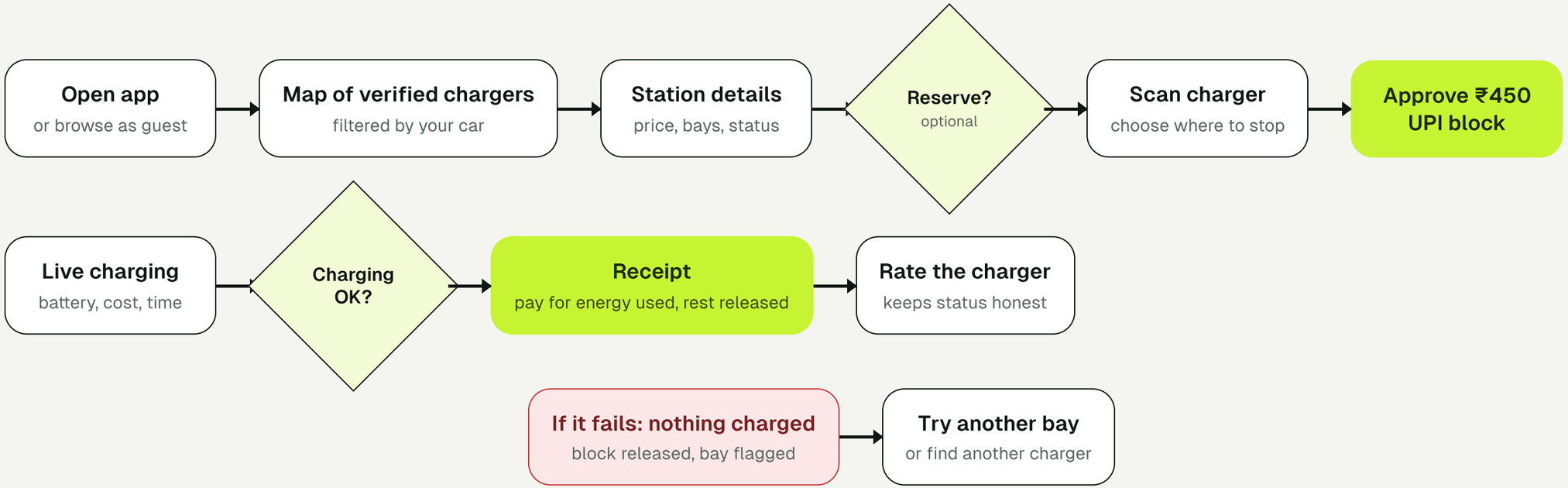
PAIN POINTS

- Worries about finding a working charger on the highway
- Finds wallet top-ups and refunds confusing
- Wants to plan his stops before he leaves

Where drivers get stuck, and what changes

	DOING	WORRY	DESIGN ANSWER
Find	Opens the app before leaving	“Is that charger really working?”	**Verified working, from the last successful charge**
Decide	Compares chargers nearby	“What will this cost my car?”	**Full price with GST and an estimate for your car**
Reserve	Holds a slot	“Will it be taken when I arrive?”	**Free reservation with a grace period**
Charge	Plugs in and starts	“Do I need another wallet?”	**UPI block per session, no top-up**
Pay	Unplugs and leaves	“Was I charged correctly?”	**Receipt with GST split and auto release**
If it fails	Session drops	“Where is my money?”	**Nothing charged, block released, bay flagged**

The charging flow



How the design answers it

01

Verified working, not just “available”

Each charger shows its last successful charge and success rate. Drivers confirm it after every session.

02

One app across networks

Chargers from every network on one map; browse without an account.

03

No wallets

Pay per session by UPI: approve a ₹450 block, pay only for energy used, the rest is released automatically.

04

The full price, upfront

Price per kWh with GST, plus cost and time for your car, before you plug in.

05

Your car sets the filters

Add your car once and only see connectors that fit.

06

A plan for when it fails

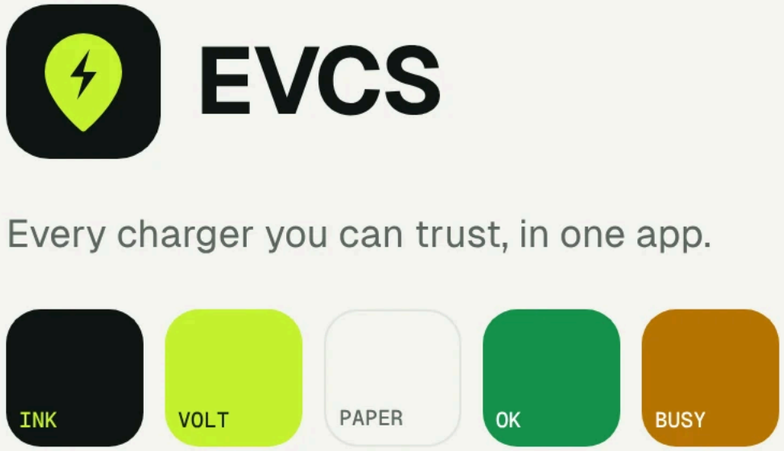
Nothing charged, block released, bay flagged for other drivers.

Mood board


Calm, even at 11 pm on a highway.
Designed for a driver in a hurry, in a car, often at night. Dark ink, one bright accent, nothing else competing.
MOOD · NIGHT
MOOD · MOVEMENT, NOT RANGE ANXIETY
THREE WORDS
Trustworthy
Green only when it really works.
Electric
One bright volt accent, used with care.
Calm
Plain words, big numbers, no clutter.
EV
Like a road sign: high contrast, readable at a glance.
TYPE · GEIST
Aa
₹19.50
Regular to Bold
KIT · STATUS FIRST
₹19.5
₹21
₹17
Verified working
Busy · 14 min
Offline
Start charging
Reserve a slot
Ink
#0F1714
The city at night
Volt
#C6F432
A charger that's ready
Paper
#F4F5F0
Daylight, calm
Working
#16924C
Verified, go
Busy
#B87503
Wait a little
Offline
#C9CEC7
Don't go
NUMBERS PEOPLE CHECK
64%
14 min left · ₹318 so far
BIG, GLANCEABLE, LIVE

A new identity

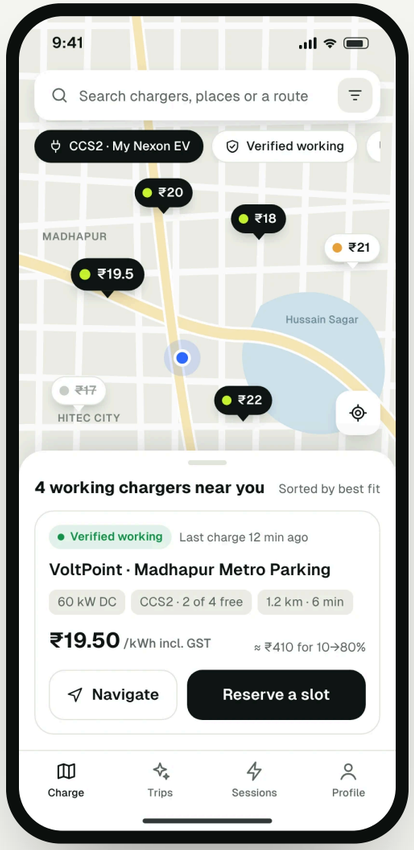
The 2019 logo was redrawn on a proper grid. The map pin and lightning bolt stay, with dark ink and a bright volt green, so status colours stand out on every screen.



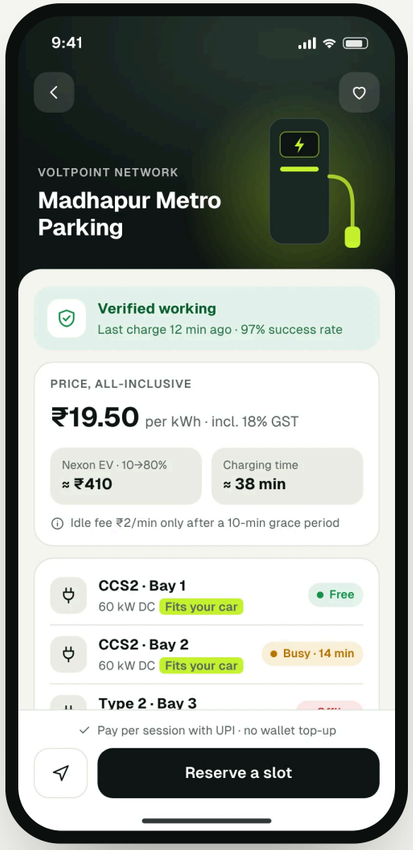


Find a charger you can trust

Pins show price and live status. The station page answers: does it work now, what will it cost my car, which bay fits?



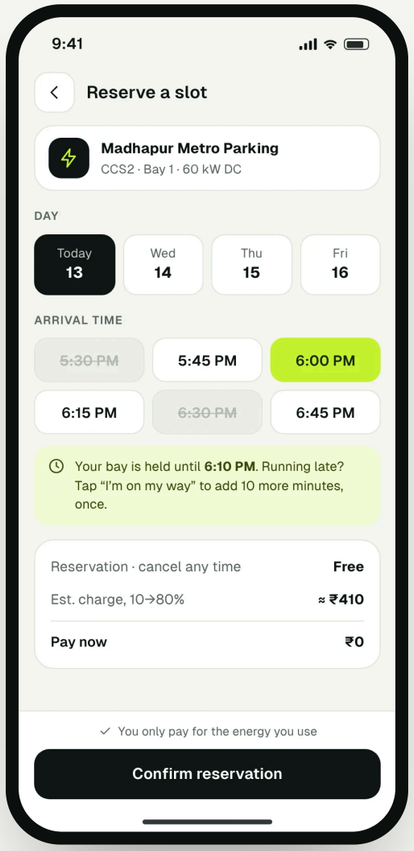
Map



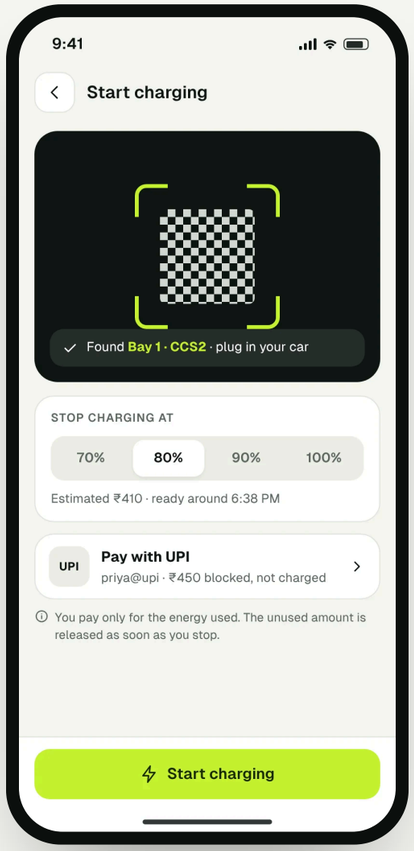
Station details

Reserve, charge and pay, without a wallet

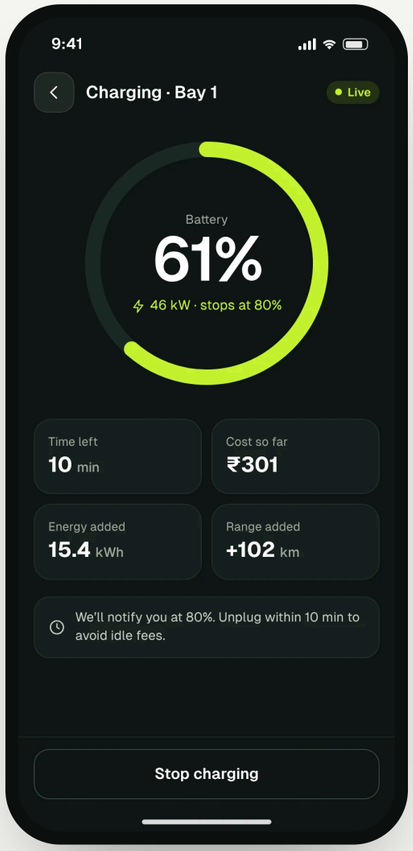
A free reservation, a QR scan to start, a UPI block instead of a top-up, and a receipt that splits out GST.



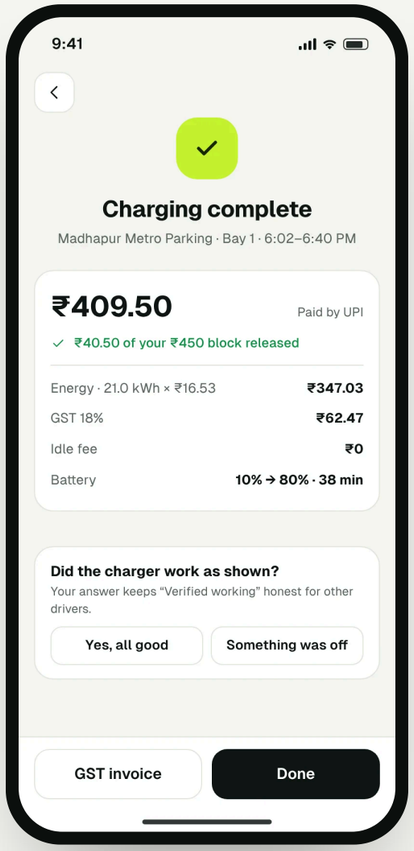
Reserve a slot



Start and pay



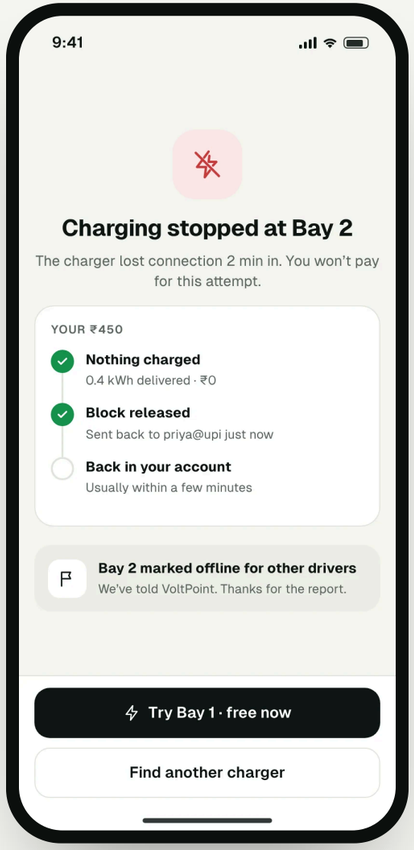
Live charging



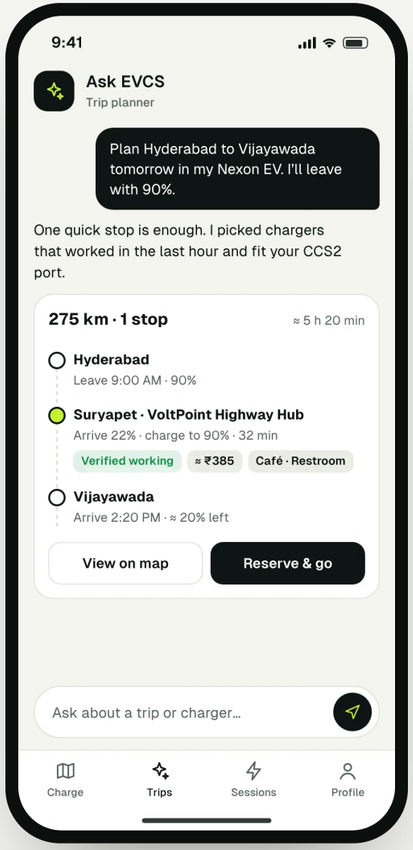
Receipt

Designed for the bad day too

When a session fails, nothing is charged and a working bay is one tap away. Ask EVCS plans road trips using only verified chargers.



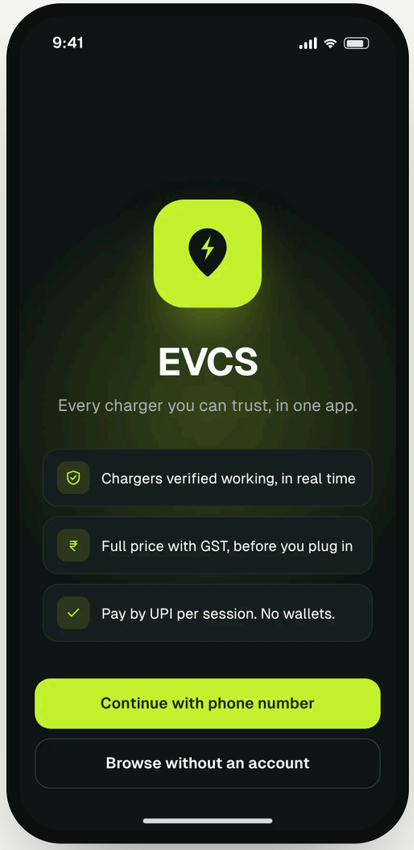
Failed session



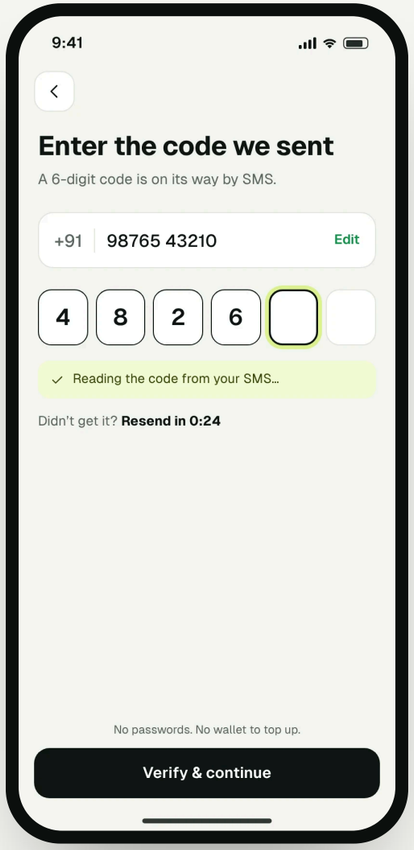
Ask EVCS trip planner

Three steps, or none

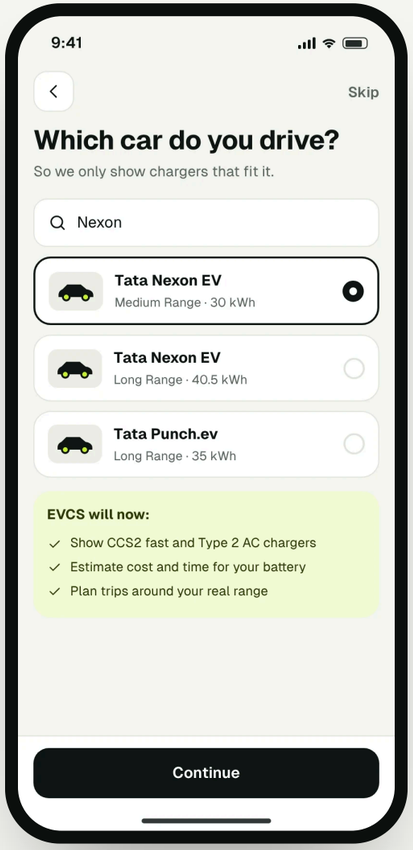
A phone number, a code read from SMS, and your car. Or skip it and browse the map.



Welcome



Phone sign-in



Add your car

11 · OUTCOME

A concept, built on real problems

In 2019, testing with 5 users found the screens easy to understand and the reservation option useful, and asked for prices on every station. The 2026 redesign closes that gap and answers the problems drivers report today.

Next, I'd test the prototype with EV drivers in Hyderabad, starting with cab drivers, and measure how quickly they find a working charger and whether they trust the "Verified working" badge.

Thank you

Clickable prototype: shivashankerbantu.github.io/ev-prototype
shiva.bantu8272@gmail.com